

PROFESSIONAL SUMMARY:

- Highly experienced Senior Data Engineer with five years of expertise in developing and optimizing robust ETL pipelines using PySpark and Databricks.
- Adept at implementing CI/CD pipelines for Databricks notebooks and jobs, ensuring streamlined deployment and operational efficiency.
- Specialized in leveraging Delta Lake for establishing ACID transactions, enhancing data reliability, and maintaining data integrity across large datasets.
- Proven ability to optimize data ingestion strategies from diverse sources, including APIs, streaming platforms, and traditional batch processes.
- Expert in ensuring strict compliance with evolving data governance and security standards within complex data ecosystems.
- Skilled in fostering collaborative relationships with data engineers and scientists to support critical data pipelines and machine learning workflows.
- Proficient in conducting thorough ETL and comprehensive data quality analysis using advanced Python and Databricks functionalities.
- Committed to consistently meeting stringent data governance and quality assurance standards throughout the entire data lifecycle.
- Experienced in organizing and leading productive meetings, including meticulous scheduling, agenda drafting, and detailed minute delivery.
- Capable of providing and archiving all required documentation, diligently tracking action items, and ensuring timely follow-up for project completion.
- Adept at summarizing complex information and presenting insightful reports to technical teams, offering clear and actionable recommendations.
- Strong background in Agile SDLC, sprint planning, and defect tracking using JIRA, ensuring efficient and collaborative project delivery.

WORK EXPERIENCE:

Senior Data Engineer @ Optum | Eden Prairie, MN

Aug 2024 – Present

- Designed and meticulously implemented scalable data pipelines on AWS, specifically tailored to support intricate healthcare analytics and operational reporting needs.
- Developed and significantly optimized PySpark applications on Amazon EMR, processing extensive volumes of critical claims and eligibility data with high efficiency.
- Constructed robust ETL workflows using AWS Glue, effectively ingesting vital data from on-prem Oracle systems directly into secure Amazon S3 data lakes.
- Implemented advanced data transformations utilizing Spark SQL and enhanced job performance through strategic partitioning and intelligent bucketing strategies.
- Integrated diverse healthcare data streams from both REST APIs and various flat file formats, including comprehensive CSV and complex JSON structures.
- Stored meticulously curated datasets in optimized Parquet format, substantially improving query performance and significantly boosting overall storage efficiency.
- Implemented rigorous data quality checks and robust reconciliation frameworks, ensuring HIPAA-compliant data processing and maintaining stringent data accuracy.
- Secured sensitive healthcare data comprehensively using robust IAM policies, implementing encryption at rest and also ensuring secure data in transit.
- Enabled seamless data consumption through direct Athena queries and interactive Tableau dashboards, providing valuable insights for analytics teams.
- Orchestrated complex batch workflows effectively using Apache Airflow, incorporating advanced alerting mechanisms and comprehensive monitoring capabilities.
- Implemented centralized logging solutions and sophisticated exception handling routines, ensuring high reliability for critical production pipelines.
- Actively participated in all Agile Scrum ceremonies and committed to sprint-based delivery cycles, meticulously managing tasks using JIRA.

- Collaborated closely with data scientists to optimize data accessibility and pipeline efficiency, supporting advanced machine learning initiatives.
- Developed and deployed robust CI/CD pipelines to automate the integration and delivery of Databricks notebooks and jobs.
- Ensured strict adherence to data governance and security standards while handling highly sensitive healthcare information.

Technologies Used: AWS (S3, EMR, Glue, Athena, Redshift), Spark, PySpark, Airflow, Oracle, Parquet, Tableau, GitHub, Jenkins, Databricks, Delta Lake

Data Engineer @ Wells Fargo | San Francisco, CA

Apr 2022 – Jun 2023

- Designed and developed highly scalable cloud-based data pipelines on Azure, specifically for critical banking and intricate risk analytics.
- Implemented efficient ETL workflows leveraging Azure Databricks and Spark to effectively process extensive transactional and valuable customer data.
- Built reliable data ingestion pipelines using Azure Data Factory, loading data from SQL Server and various APIs into ADLS Gen2.
- Developed complex data transformations using PySpark and meticulously optimized Spark jobs for superior performance and cost efficiency.
- Implemented sophisticated analytical data models utilizing Azure Synapse Analytics, enhancing data insights and reporting capabilities.
- Successfully migrated legacy on-prem Hadoop workloads to advanced Azure cloud platforms, ensuring seamless transition and improved scalability.
- Implemented stringent row-level security and comprehensive column masking techniques to meticulously meet evolving financial compliance requirements.
- Enabled downstream analytics by creating dynamic Power BI dashboards and performing intricate Synapse SQL queries for business stakeholders.
- Automated critical build and deployment pipelines using Jenkins, ensuring rapid and consistent delivery of data solutions.
- Managed source code diligently using GitHub, strictly adhering to established standard branching strategies for version control.
- Performed extensive data validation, rigorous unit testing, and meticulous reconciliation checks to maintain high data accuracy.
- Participated actively in Agile sprint planning sessions, detailed reviews, and constructive retrospectives, managing tasks with JIRA.
- Collaborated with data scientists to optimize data models and ensure robust data quality for their machine learning experiments.
- Contributed to implementing CI/CD practices for Databricks notebooks and jobs, improving development lifecycle efficiency.
- Ensured strict data governance and quality assurance standards were met throughout the entire data processing lifecycle.

Technologies Used: Azure (ADLS Gen2, ADF, Databricks, Synapse), Spark, PySpark, SQL Server, Power BI, GitHub, Jenkins, Delta Lake

Junior Data Engineer @ Big Lots | Columbus, OH

Nov 2019 – Mar 2022

- Designed and actively supported foundational data pipelines for retail and merchandising analytics, ensuring data availability for critical business functions.
- Developed efficient ETL workflows utilizing Informatica to ingest data from both Oracle databases and various flat file sources.
- Performed complex data transformations using advanced SQL and PL/SQL scripting, preparing data for reporting and analysis.
- Built and diligently maintained dimensional data models, providing robust support for comprehensive sales and inventory reporting.
- Optimized SQL queries significantly to improve overall reporting performance and accelerate batch processing efficiency.
- Implemented thorough data validation and essential reconciliation checks, consistently ensuring high data accuracy and reliability.

- Supported the creation and maintenance of reporting datasets, which were actively consumed by various downstream BI teams.
- Managed version control effectively using GitHub, strictly adhering to established standard deployment processes and protocols.
- Assisted senior engineers in troubleshooting and resolving data pipeline issues, ensuring continuous operation and data flow.
- Contributed to the documentation of data processes and system architecture, using tools like Confluence for knowledge sharing.
- Participated in team meetings to discuss project progress and address technical challenges, fostering a collaborative environment.
- Gained foundational experience with data quality analysis and initial data governance principles in a retail environment.
- Assisted in optimizing data ingestion processes from various batch sources to ensure timely data availability.
- Learned about ensuring compliance with basic data security standards for retail operational data.
- Developed foundational skills in ETL development, data modeling, and metadata management for business intelligence initiatives.

Technologies Used: Oracle, Informatica, SQL, PL/SQL, GitHub, Jenkins, Databricks (exposure), Delta Lake (exposure)

TECHNICAL SKILLS:

- **Programming Languages:** Python, SQL, PySpark, Scala
- **Big Data Technologies:** Databricks, Spark, Delta Lake, Snowflake, Hadoop, Hive
- **Cloud Platforms:** AWS (S3, EMR, Glue, Athena, Redshift, Step Functions), Azure (ADLS Gen2, ADF, Databricks, Synapse)
- **Data Engineering Tools:** Apache Airflow, Kafka, ETL, Informatica, Oracle, MySQL, PostgreSQL, SQL Server, MongoDB
- **CI/CD & DevOps:** Jenkins, Docker, GitHub, Git
- **Data Governance & Quality:** Data Governance, Quality Assurance, ACID Transactions, Data Reliability
- **Business Intelligence:** Tableau, Power BI, Azure Analysis Services
- **Project Management & Collaboration:** JIRA, Confluence, Agile (Scrum), Lucidchart

EDUCATION:

Master of Science in Computer Science @ University of Central Missouri